

JIZAN, Saudi Arabia

WMO: 411400

Lat: 16.8964N

Lon: 42.5837E

Elev: 20

StdP: 14.69

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 71.3	(c) 72.7	(d) 60.7	(e) 79.4	(f) 80.2	(g) 62.8	(h) 85.6	(i) 79.6	(j) 19.8	(k) 83.5	(l) 18.3	(m) 83.4	(n) 3.9	(o) 110	(p) 0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.7	(c) 100.9	(d) 82.2	(e) 100.3	(f) 82.5	(g) 98.9	(h) 82.9	(i) 86.8	(j) 96.8	(k) 86.1	(l) 96.2	(m) 85.4	(n) 95.6	(o) 12.3	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
84.4	180.5	94.6	83.8	177.3	94.3	82.7	170.5	93.9	51.7	95.9	50.8	96.4	50.2	96.2	94.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.4	(b) 18.0	(c) 16.1	(e) 66.9	(f) 104.9	(g) 2.8	(h) 2.0	(i) 64.9	(j) 106.4	(k) 63.3	(l) 107.6	(m) 61.8	(n) 108.7	(o) 59.8	(p) 110.2		
(a) 20.4	(b) 18.0	(c) 16.1	(e) 60.5	(f) 89.2	(g) 2.7	(h) 1.8	(i) 58.6	(j) 90.5	(k) 57.1	(l) 91.5	(m) 55.6	(n) 92.6	(o) 53.7	(p) 93.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	87.3	79.4	80.6	83.2	87.5	91.0	93.3	93.7	92.8	92.0	88.6	84.6	80.9	(6)
(7)		DBStd	5.53	1.84	1.84	2.23	2.37	2.17	2.01	2.28	2.27	1.70	1.79	2.15	1.97	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	13627	910	856	1031	1126	1272	1298	1353	1326	1261	1196	1039	959	(10)
(11)		CDD65	8152	445	436	566	676	807	848	888	861	811	731	589	494	(11)
(12)		CDH74	117074	4270	4538	6970	9790	12660	13768	14338	13673	12812	11004	7824	5427	(12)
(13)		CDH80	68434	1142	1371	2907	5525	8203	9449	9877	9215	8494	6574	3785	1892	(13)
(14)	Wind	WSAvg	7.1	6.8	6.7	6.9	6.9	6.9	7.5	8.6	8.0	7.0	6.5	6.5	6.5	(14)
(15)	Precipitation	PrecAvg	4.4	0.4	0.1	0.2	0.4	0.3	0.1	0.4	1.2	0.2	0.6	0.4	0.4	(15)
(16)		PrecMax	10.8	2.4	0.6	1.0	2.6	1.9	0.5	4.1	4.7	1.4	5.9	2.4	3.1	(16)
(17)		PrecMin	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.3	0.7	0.2	0.3	0.7	0.5	0.1	0.8	1.3	0.3	1.2	0.7	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.2	89.7	93.2	98.9	102.4	102.6	102.5	101.9	100.8	99.7	95.3	90.0	(19)
(20)			MCWB	76.6	77.6	78.9	78.6	79.8	82.6	81.6	82.7	83.0	80.5	78.3	78.2	(20)
(21)		2%	DB	87.3	88.0	91.4	96.7	100.2	100.7	100.6	100.3	100.0	97.5	93.5	89.3	(21)
(22)			MCWB	76.4	77.2	78.8	79.9	81.0	82.9	82.2	83.3	83.5	81.7	79.1	77.9	(22)
(23)		5%	DB	86.1	87.3	89.9	95.1	98.6	99.2	99.0	98.8	98.7	96.7	92.5	88.0	(23)
(24)			MCWB	75.8	76.9	78.4	80.2	81.8	83.4	82.4	83.3	83.8	82.0	79.1	77.3	(24)
(25)		10%	DB	84.7	86.0	89.2	93.5	97.0	98.5	98.4	97.9	97.5	95.3	91.3	87.1	(25)
(26)			MCWB	75.2	76.3	78.2	80.3	82.0	83.6	82.4	83.2	83.9	81.7	78.8	76.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.0	79.8	81.8	84.0	86.6	87.7	87.1	87.2	87.3	86.0	82.7	80.1	(27)
(28)			MCD B	86.5	87.4	90.1	93.7	96.6	97.5	97.2	97.0	97.1	95.4	92.2	88.2	(28)
(29)		2%	WB	77.4	78.5	80.5	82.8	85.3	86.8	85.9	86.2	86.4	84.5	81.4	79.2	(29)
(30)			MCD B	85.1	86.3	89.0	92.8	95.5	96.9	96.2	96.1	96.4	94.7	91.0	87.3	(30)
(31)		5%	WB	76.7	77.4	79.6	82.0	84.5	85.9	84.9	85.3	85.7	83.8	80.5	78.4	(31)
(32)			MCD B	84.5	85.3	88.2	92.1	94.9	96.3	95.5	95.4	95.7	94.1	90.1	86.5	(32)
(33)		10%	WB	76.0	76.8	78.9	81.2	83.7	85.2	84.2	84.6	85.1	82.7	79.8	77.4	(33)
(34)			MCD B	83.7	84.8	87.6	91.3	94.2	95.6	95.0	94.8	95.0	93.3	89.2	85.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.3	12.0	12.0	12.8	13.5	11.6	10.7	11.6	13.1	14.7	13.5	12.8	(35)
(36)			MCDBR	12.8	12.4	12.6	14.5	14.7	12.4	11.1	12.1	13.6	15.9	14.6	13.0	(36)
(37)			MCWBR	7.3	6.6	7.1	8.5	8.4	8.1	7.6	7.4	8.1	8.5	7.4	7.1	(37)
(38)		5% WB	MCDBR	12.2	12.1	11.6	12.8	12.9	11.2	10.6	11.5	13.0	14.8	13.1	12.2	(38)
(39)			MCWBR	7.5	7.1	7.2	8.6	8.3	8.0	8.1	7.7	8.4	8.6	7.5	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.500	0.554	0.588	0.591	0.649	0.733	0.795	0.775	0.682	0.566	0.529	0.497	(40)	
(41)		taud	2.070	1.926	1.853	1.860	1.730	1.565	1.504	1.535	1.676	1.894	1.982	2.068	(41)	
(42)		Ebn at Noon	249	241	238	238	221	201	189	195	213	235	238	245	(42)	
(43)		Edh at Noon	50	61	67	67	75	88	93	91	79	62	54	49	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1564	1793	2047	2216	2306	2232	2044	2019	2080	1973	1675	1554	(44)	
(45)		RadStd	79	80	93	81	55	64	83	70	43	58	55	70	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.97	+1.15	N/A	+0.50	N/A	N/A	N/A	N/A	+362	+360	(46)
(47)	Regional (2 neighbors)	+0.95	+0.81	N/A	+0.90	N/A	N/A	N/A	-22	+360	+274	(47)

Nomenclature: See separate page